E20 technical review — current owner instruction

1 November 2026 · REVIEW ONLY · This update supersedes conflicting wiring-method and purchase assumptions on the following earlier E20 sheets. Their circuit and geometry details remain proposals requiring review.

Kitchen: owner selects Romex (NM-B), subject to verification that the wiring method is permitted for the actual building and route. Final conductors, devices, locations and lengths require electrician and field review.

Internal non-kitchen outlets: after route approval and John-required carpentry, drill approved paths and install approved boxes plus EMPTY ENT. Do not pull or run wire until inspection AND release. This is an owner instruction, not a claim that a particular inspection is universally required by code or has passed. ENT listing, location, protection, fittings, support, bends, fill, boxes and later conductor type need trade/AHJ review. Kitchen is a separate method; non-outlet HVAC and lighting require their own method review.

Laundry: internal execution includes the owner-selected LG WKG101HVA gas WashTower. LAUN-W and LAUN-D on the following E20 proposal are preliminary; reconcile against the actual issued schedule to avoid missing or duplicate branches. The outside-facing P2 kitchen/HVAC schedule had no laundry rows. Photograph actual labels and complete the all-served-load and feeder review. The 20-space / 18 single-pole scenario already includes both proposed laundry rows; it does not prove final fit or amperage capacity.

Takeoff: old 75 ft 12/2 washer, 55 ft 14/2 dryer and other non-kitchen NM-B lengths on later ET-03 are historical allowances only. They are not measured ENT lengths or a purchase list. Use the revised separate EN/ES takeoff and laundry execution package. No purchase, inspection approval, wire release, energization or construction release is implied.

Eaton 20-space selection and framing interface

Electrical technical handoff to the existing interior lead | Owner choice incorporated; physical fit and quantities remain review items

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Selection: Eaton **BRP20L125G**, 125A main-lug-only, **20 full-size spaces**. The current source has **18 proposed single-pole circuits**, leaving two positions in the proposed arrangement. The required **100A copper feeder** is retained as the working owner requirement. **Purchase hold, 1 Nov:** panel purchase paused until Monday 2 Nov; no charge/payment request today and no automatic Monday purchase. Current quote and payment approval are required then.

Item	Verified / proposed basis	Release boundary
Model and contents	Eaton catalog BRP20L125G; cover and ground bar included. Matching cover BRPX2A20L1F. Long string BRP20L125G125AMLO retained only as descriptive source history.	125A is the panel rating, not an included 125A main breaker. MLO supplied through reviewed upstream protection.
Can / cover	SKU product dimensions 21 x 14.31 x 3.88 in; approximate X2 outline 21.10 x 14.30 x 3.80. Cover outside: 22 H x 15.3 W x 1 D in.	Distinct sources retained. None is a rough opening. Actual can AND matching trim must be in hand before framing.
D04 placement	DINING WEST ENTRY INFILL face; panel door toward dining. Proposed center x = W/4 + 1 in and 65 in AFF.	Numeric W and confirmed mounting reference unavailable. No guessed wall width, stud cuts or finished-floor offset.
Feeder / terminals	100A working requirement: #3 Cu THWN-2 phase conductors and full neutral; #8 Cu EGC baseline. Published lug range #6-2/0 Cu/Al includes #3.	Verify 75C terminations, actual label/torque, upstream breaker terminal range, thermal corrections, fill, grounding and completed load. No feeder length released.

Exact D04 datum and envelope check

W: finished clear span from the outside edge of the rightmost window casing to the right inside corner of the west wall, viewed facing D04. The transom doorway on the perpendicular wall is excluded. R2 proposes two upper frames with B = (W - 12)/2; W is still missing. If 65 AFF denotes the *cover center*, the 22-in cover would span 54-76 AFF. This conditional arithmetic does not establish the box, handle or rough-opening datum.

Interior lead to overlay actual cover, open-door/removal envelope and moulding. Confirm stud conditions, finish thickness, electrical working space and handle accessibility before mark/cut release. The older approximate 12-in living-face location and 40/42-space drafts are superseded for this decision.

Sources: Eaton exact panel sheet | Eaton cover match | Eaton cover dimensions | D04 coordination note (1 Nov), derived from R2 pp3-4. Current sources and hashes retained with this package.

Provisional 20-space Subpanel schedule

120/240V single-phase, 3-wire plus equipment ground | MLO | Ten source kitchen/HVAC circuits + eight whole-unit proposals

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Positions below are a review allocation, not the original source circuit numbers. All 18 listed proposals are 120V / single-pole; **18 x 1 = 18 spaces**. Slots 17 and 19 are left unassigned on one side for an adjacent two-pole position if needed. Verify factory numbering and bus sequence before installation; phase balancing requires completed loads.

Left slot / leg*	Circuit / load	Breaker / protection	Right slot / leg*	Circuit / load	Breaker / protection
1 / A	**SA-A** - East countertop, E1/E2	20A / 1P / DF	2 / A	**SA-B** - South countertop, S1/S2	20A / 1P / DF
3 / B	**SA-C** - West countertop, W1/W2	20A / 1P / DF	4 / B	**REF** - Refrigerator	20A / 1P / DF
5 / A	**DW** - Dishwasher	15A / 1P / DF	6 / A	**GD** - Switched disposal	15A / 1P / DF
7 / B	**RG** - Gas range	15A / 1P / DF	8 / B	**HD** - Hardwired hood	15A / 1P / AFCI
9 / A	**LT** - Kitchen + task lights	15A / 1P / AFCI	10 / A	**HVAC** - Furnace + disconnect	15A / 1P / Confirm
11 / B	**LIV** - Living room + hallway	20A / 1P / AFCI	12 / B	**DIN** - Dining room outlets	20A / 1P / AFCI - review
13 / A	**BED** - Bedroom outlets	20A / 1P / AFCI	14 / A	**BATH-R** - Bathroom receptacles only	20A / 1P / GFCI devices
15 / B	**BATH-L** - Ceiling + vanity lighting	15A / 1P / Confirm	16 / B	**LAUN-W** - LG washer	20A / 1P / DF
17 / A	**SPACE** - unassigned; no breaker	--	18 / A	**LAUN-D** - LG gas dryer	15A / 1P / DF
19 / B	**SPACE** - unassigned; no breaker	--	20 / B	**ATTIC** - Service light + receptacle	15A / 1P / DF

Capacity accounting: 20 physical spaces - 18 proposed branch poles = 2 unassigned. The upstream feeder breaker does not occupy this MLO subpanel. No tandem breakers are assumed. A future two-pole load, panel-mounted surge device or a change to main/backfeed configuration can consume the two positions and may require a revised arrangement. Forty advertised circuits does not mean forty full-size spaces.

Protection key: DF = combined AFCI/GFCI function; AFCI = arc-fault protection. Protection and breaker ampere values are inherited proposals from C1, not a release. DIN small-appliance classification/protection, BATH-L controls/fan, BATH-R toilet load, furnace model and all final breaker catalog selections remain to verify. Gas dryer is the recorded 120V LG proposal; no electric dryer or 240V range has been substituted.

Source: whole-unit package-data.json (27 Oct source, 29 Oct retained snapshot); all 18 rows preserved. Later 20-space owner selection supersedes its >=40-space material line. *Leg sequence illustrated for review; physical product label governs. This is a space reconciliation, not a completed whole-unit feeder load calculation.

Romex route register and quantity gaps

Every current source circuit included | All lengths below are HISTORICAL ALLOWANCES, not measured routes or order quantities

ID	Cable proposal + ground	Route to measure / confirm	Old allowance ft	Measured ft / stock
SA-A	12/2 + G Cu NM-B	D04 Subpanel to E1/E2; same-partition direct route intent; remeasure after panel setout.	45	TBD / TBD
SA-B	12/2 + G Cu NM-B	Subpanel up to attic, across to SOUTH, down to S1/S2; counter interconnection.	65	TBD / TBD
SA-C	12/2 + G Cu NM-B	Subpanel up to attic, across to WEST, down to W1/W2; range separates counters.	85	TBD / TBD
REF	12/2 + G Cu NM-B	Subpanel to Haier outlet on kitchen EAST partition; direct-wall route intent.	35	TBD / TBD
DW	14/2 + G Cu NM-B	Subpanel via attic to accessible sink-base dishwasher receptacle.	25	TBD / TBD
GD	14/2 + G Cu NM-B	Subpanel via attic to motor-rated sink switch, then switched sink-base receptacle.	35	TBD / TBD
RG	14/2 + G Cu NM-B	Subpanel via attic to WEST Viking utility zone; verify exact outlet location.	45	TBD / TBD
HD	14/2 + G Cu NM-B	Subpanel via attic to WEST Zephyr connection; final appliance whip per listing.	50	TBD / TBD
LT	14/2 + 14/3 + G Cu NM-B	Up/across/down to entry controls; separate MAIN and PEND returns; P1 to P2; UC driver input.	65	TBD / TBD
HVAC	14/2 + G Cu NM-B	Subpanel to attic furnace disconnect; exact Trane/model interface and whip pending.	60	TBD / TBD
LIV	12/2 + G Cu NM-B	Subpanel to living/hall devices; final wall marks and interconnection route unmeasured.	100	TBD / TBD
DIN	12/2 + G Cu NM-B	Subpanel to dining devices; revised D04 marks and small-appliance/protection review pending.	85	TBD / TBD
BED	12/2 + G Cu NM-B	Subpanel to bedroom devices; final furniture/outlet layout and route unmeasured.	105	TBD / TBD
BATH-R	12/2 + G Cu NM-B	Subpanel to bathroom vanity/toilet receptacles; final device and toilet load pending.	65	TBD / TBD
BATH-L	14/2 + G Cu NM-B	Subpanel to bathroom controls, ceiling and vanity lights; fan/control topology pending.	55	TBD / TBD
LAUN-W	12/2 + G Cu NM-B	Subpanel to LG WKG101HVA washer receptacle; final route/manual check pending.	75	TBD / TBD
LAUN-D	14/2 + G Cu NM-B	Subpanel to LG WKG101HVA gas-dryer receptacle; final route/manual check pending.	55	TBD / TBD
ATTIC	14/2 + G Cu NM-B	Subpanel to attic service receptacle, access switch and light; independent of furnace supply.	60	TBD / TBD

Ground/count: 12/2 has two insulated #12 Cu conductors plus #12 equipment ground; 14/2 has two insulated #14 plus #14 ground; 14/3 has three insulated #14 plus #14 ground. /2 or /3 excludes the ground. Supplemental 14/3 lighting/control route is not located; the older 70-ft allowance is separate from the LT 14/2 allowance.

Method gate for every row: NM-B only for verified suitable dry locations and permitted building/route classification. Verify SF 334.10/.12, protection from damage, supports, grouping/thermal adjustment and firestopping. Furnace/hood final connection methods follow their listings. Exterior feeder uses separate rated conductors in conduit, not NM-B. No route is certified permitted solely by this table.

Cable procurement reconciliation

Source-backed roll options and exact outstanding inputs | Parent retains checkout; no quantity released by this sheet

Cable / manufacturer roll	Prior quantity calculation (C1)	Current purchase status
12/2 + ground - Southwire 28828255, 250 ft	Nine source circuit allowances total 660 ft. Prior 10% combined handling/tails = 726 ft; prior proposal 3 x 250 = 750 ft.	Measured segments, tails, accepted routes and usable stock unknown. Deficit and roll count NOT RELEASED.
14/2 + ground - Southwire 28827455, 250 ft	Nine source circuit allowances total 450 ft. Prior 10% combined handling/tails = 495 ft; prior proposal 2 x 250 = 500 ft.	Same quantity hold. Confirm 15A circuit proposals, furnace interface and final switching before grouping rolls.
14/3 + ground - Southwire 63946928, 100 ft	70 ft supplemental control allowance; prior 10% = 77 ft; prior proposal 1 x 100 ft.	Exact switched/control conductors and route unresolved. Do not add a roll merely because it appears in the old budget.
100A feeder - not Romex	Old 75-ft physical route / 85-ft each-wire allowance and 90-ft conduit stock were unmeasured. Old #4 / 80A allowance conflicts with O2.	Working requirement 100A; #3 Cu fits selected panel main lugs. No wire length, raceway trade size or quantity released.

Inputs that release quantities

Missing input	Quantity affected / next action
D04 actual panel setout; approved device marks; attic/wall route segments and vertical drops	All 18 home runs and downstream segments. Measure along accepted paths; wall dimensions alone are insufficient.
Final MAIN/PEND/UC and bathroom control wiring; furnace and hood connection method	14/2 versus 14/3 allocation, control tails and any separate appliance connection materials. Avoid double-counting a shared physical route as one cable.
Usable existing stock by cable marking, length and continuous piece	Every purchase deficit. Unknown stock is not zero stock; confirm conductor count and condition before deducting.
Termination tails, cutting allowance and continuous cut list	For each type: required stock = measured cable lengths + explicit tails + accepted waste; deficit = required stock minus usable stock (minimum zero). Select rolls against the cut list, not total length alone.

NM-B uses its 60C ampacity basis; ordinary 14 Cu / 15A and 12 Cu / 20A proposals remain subject to the applicable installation checks. At verified 75C terminals, Southwire lists #3 Cu THWN-2 at 100A and #4 at 85A before adjustments. Physical lug fit does not establish ampacity. The prior larger-panel #1 minimum lug issue is superseded by the exact selected model.

Primary references: 12/2 roll | 14/2 roll | 14/3 roll | THWN-2 ampacity table | SF code edition | SF NM cable amendments. Read alongside the saved C1 source and editable CSV/JSON. This supplement does not revise permit scope or approve installation.